Three Views of a Limit

Precalculus / Introduction to Limits

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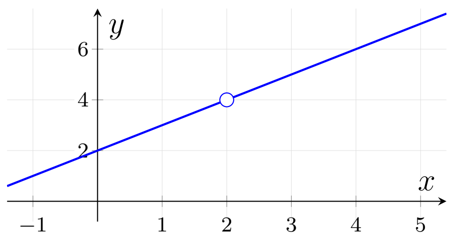
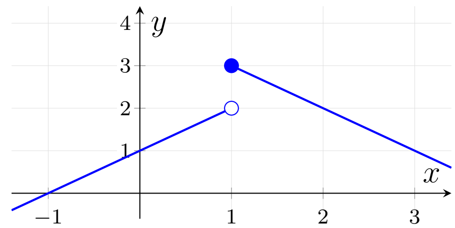
reading a limit from a table, from a graph, from algebra, and as the limit of partial sums

A limit describes where a function is *headed* as the input approaches a value — never what happens *at* that value. This sheet asks the same question through four lenses: a table of values, a graph, algebra, and the partial sums of a series. After the warm-up the four are interleaved, so decide which tool the problem is handing you before you start computing.

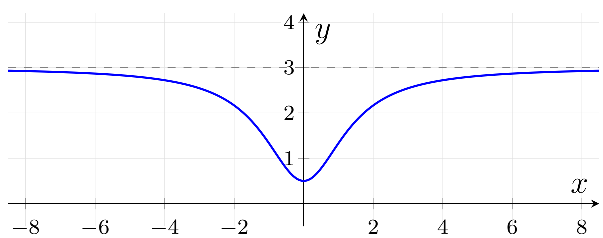
Graphs for problems 3, 7 and 11. The three graphs this sheet asks you to read are printed together here, each named. Every problem that needs one says which graph it means — *no graph belongs to any other problem.*



Graph A — the function g (problem 3).



Graph B — the function h (problem 7).



Graph C — the function p, with its horizontal asymptote dashed (problem 11).

1. The table below has already been computed for $f(x) = \dfrac{x^2 - 9}{x - 3}$.

x	2.9	2.99	2.999	3	3.001	3.01	3.1
$f(x)$	5.9	5.99	5.999	undefined	6.001	6.01	6.1

What does the table say $\lim_{x \to 3} f(x)$ is? Say in one sentence why "undefined at $x = 3$" does not stop the limit from existing.

Answer: ________________

2. Evaluate algebraically: $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$. Direct substitution gives $\frac{0}{0}$, so rewrite before you substitute.

Answer: ________________

3. **Use Graph A** (printed with the directions), which shows $g(x) = \dfrac{x^2 - 4}{x - 2}$. Its open circle marks the one input where g is not defined.
Read $\lim_{x \to 2} g(x)$ from the graph, and say in one sentence why the open circle does not change the answer.

Answer: ________________

4. Write out and evaluate the partial sum $\displaystyle\sum_{n=1}^{5} (2n - 1)$.

Answer: ________________

5. Let $f(x) = \dfrac{\sqrt{x+1}-1}{x}$. Complete the table, rounding each value to four decimal places.

x	-0.01	0	0.01
$f(x)$		undefined	

Then state the exact value of $\lim_{x\to 0} f(x)$ as a fraction. Write the two table entries and the limit on your answer line.

Answer: ____________________

6. Evaluate algebraically: $\lim_{x\to 2} \dfrac{x^3-8}{x^2-4}$. (Hint: $a^3 - b^3 = (a-b)(a^2+ab+b^2)$ and $a^2 - b^2 = (a-b)(a+b)$.)

Answer: ____________________

7. **Use Graph B** (printed with the directions), the piecewise function h. Its solid dot marks the value of $h(1)$; its open circle marks a point the graph approaches but does not reach.
(a) $\lim_{x\to 1^-} h(x)$ **(b)** $\lim_{x\to 1^+} h(x)$ **(c)** Does $\lim_{x\to 1} h(x)$ exist? Justify your answer using your values from (a) and (b).

Answer: ________________

8. Evaluate the infinite geometric series $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$, and state the partial-sum limit it represents.

Answer: ________________

9. Evaluate algebraically: $\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$ by multiplying by the conjugate of the numerator.

Answer: ________________

10. Let $r(x) = \dfrac{2x+3}{x-1}$. Complete the table, rounding each value to four decimal places.

x	100	1000
$r(x)$		

Then state $\lim_{x\to\infty} r(x)$. Write both table entries and the limit on your answer line.

Answer: ________________

11. **Use Graph C** (printed with the directions), which shows $p(x) = \dfrac{3x^2+1}{x^2+2}$ with its horizontal asymptote dashed.
Read $\lim_{x\to\infty} p(x)$ from the graph, then confirm it algebraically by dividing the numerator and denominator by x^2.

Answer: __________________

12. **Synthesis.** Consider $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$.

(a) Compute the fifth partial sum $S_5 = \sum_{n=1}^{5} \left(\frac{1}{2}\right)^n$ as an exact fraction.
(b) Find the sum of the infinite series.
(c) A classmate says "an infinite sum just means you keep adding forever". Using your answers to (a) and (b), write two or three sentences explaining what the sum of an infinite series really is, and how the numerical, graphical and algebraic views of this series agree.

Answer: __________________